

SAT

Conversational E-commerce Review Analysis System Evaluation Report

Frozen test set: 504 queries, 8 categories

Systems compared: A: LLM-only, B: Vanilla RAG, E: Full SAT

Executive Summary

What this measures

This report evaluates SAT's full quad-layer pipeline (System E) against two baselines using the SAME underlying fine-tuned model weights, so differences are attributable to the retrieval/reasoning architecture, not to model tuning. All numbers come from the frozen test split (eval_test.json), never used to tune any threshold in this system.

The headline hallucination number

System A: 50.0% hallucination rate on questions with zero real evidence. System B: 42.9% hallucination rate on questions with zero real evidence. System E: 30.4% hallucination rate on questions with zero real evidence.

How hallucination was measured

The primary metric is fully objective, not a judge's opinion: for questions where the review corpus genuinely contains zero relevant evidence (verified independently via keyword matching, not the system's own retrieval), any confident answer is by construction a fabrication. A secondary, disclosed-as-imperfect LLM-judge check on answerable questions provides a second signal (Section 4).

Read next

Section 2: retrieval quality. Section 3: the core hallucination results. Section 4: judge-based grounding check. Section 5: contradiction handling & clarification. Section 6: efficiency. Section 7: findings and what should be fixed next. Section 8: limitations.

1. Methodology

Three systems, one model

System A (LLM-only) answers from parametric knowledge with zero retrieval. System B (vanilla RAG) retrieves the top-8 sentences by raw BLAIR-dense similarity with no aspect-aware pruning threshold and answers in one shot, no persona/abstention logic. System E is the full pipeline: Doorman (intent/aspect/persona parsing) -> Librarian (hybrid BM25+BLAIR retrieval, sentence-level pruning, evidence-sufficiency gate) -> Analyst (positive/negative/neutral evidence structuring, contradiction flagging) -> Spokesperson (evidence-constrained generation or deterministic abstention). All three use the same SFT+DPO-tuned Llama-3-8B weights.

Evaluation set construction

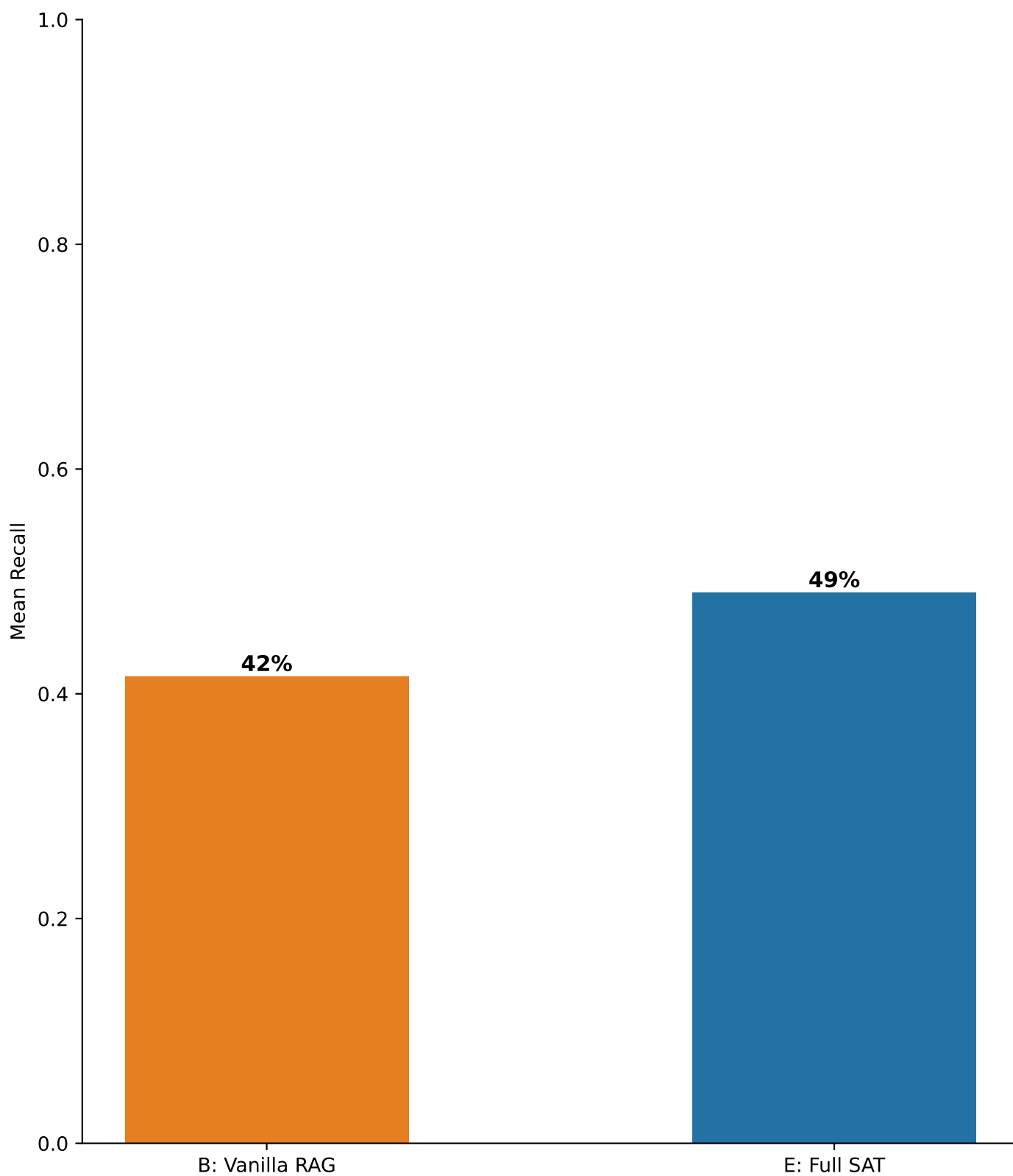
720 queries across 8 categories (aspect-specific, persona-aware, multi-aspect, overall-suitability, contradiction-heavy, unsupported-aspect, unsupported-feature, ambiguous), built entirely from products held out from training. Ground truth (which sentences are relevant, which products have zero evidence for a given aspect, which have genuinely contradictory evidence) was labeled by an independent keyword+rating-based method, deliberately NOT using the system's own semantic retrieval scores -- avoiding a circular evaluation. Split 30/70 into a dev set (used to calibrate thresholds) and this frozen test set (720 -> 504 queries, touched only for the numbers in this report).

What 'hallucination' means here

Two independent measurements. (1) Objective: on the unsupported-aspect / unsupported-feature categories, the corpus contains verifiably zero relevant evidence -- so any response that answers confidently rather than abstaining is fabricating content by construction, no judgment call required. (2) LLM-judge: a secondary check on answerable-category responses, asking whether specific claims are supported by the evidence the system actually retrieved. This is disclosed as a weaker signal -- see Section 8.

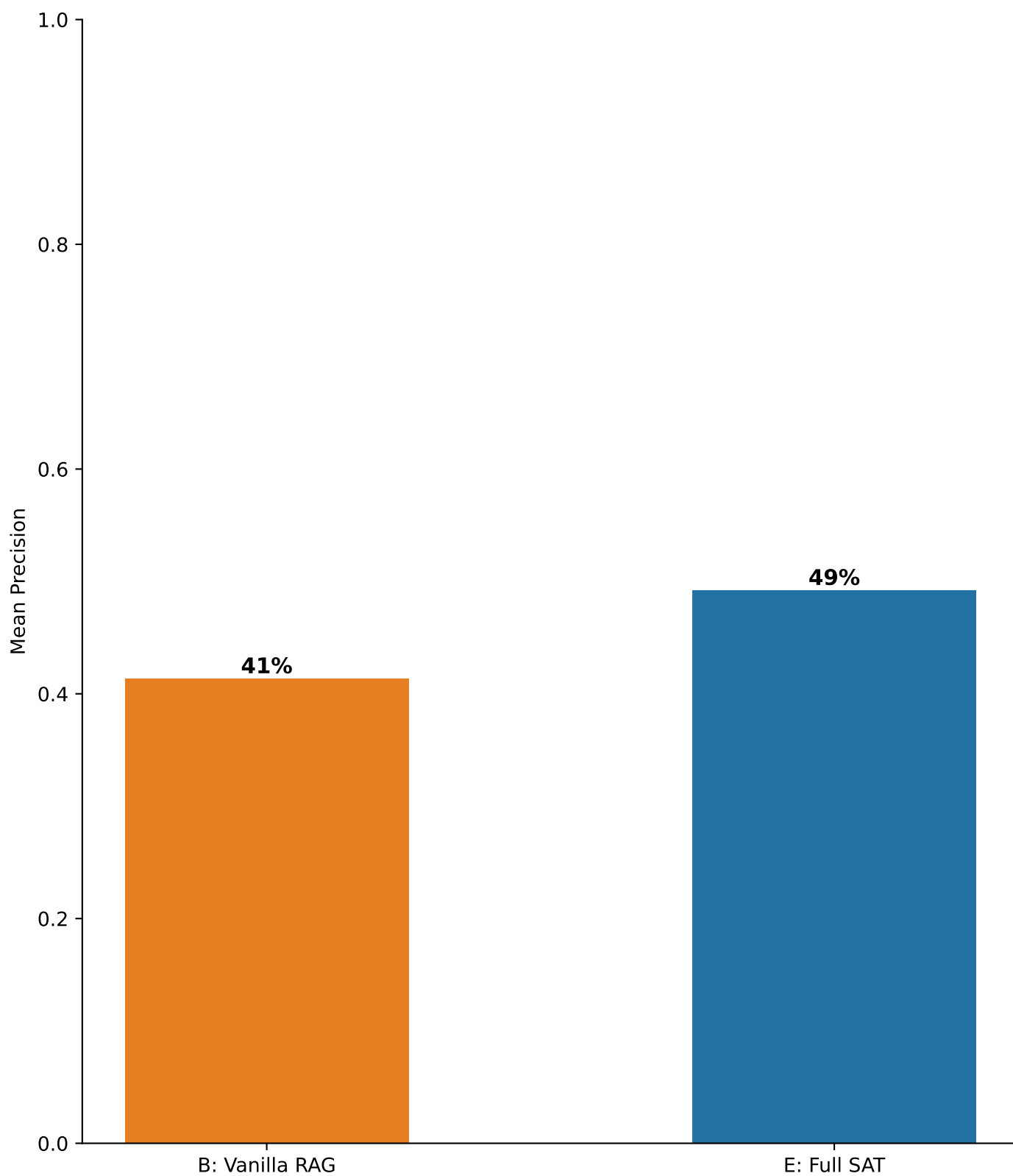
2. Retrieval Recall

Mean recall against independently-labeled gold relevant sentences (higher is better)



2b. Retrieval Precision

Mean precision against independently-labeled gold relevant sentences (higher is better)



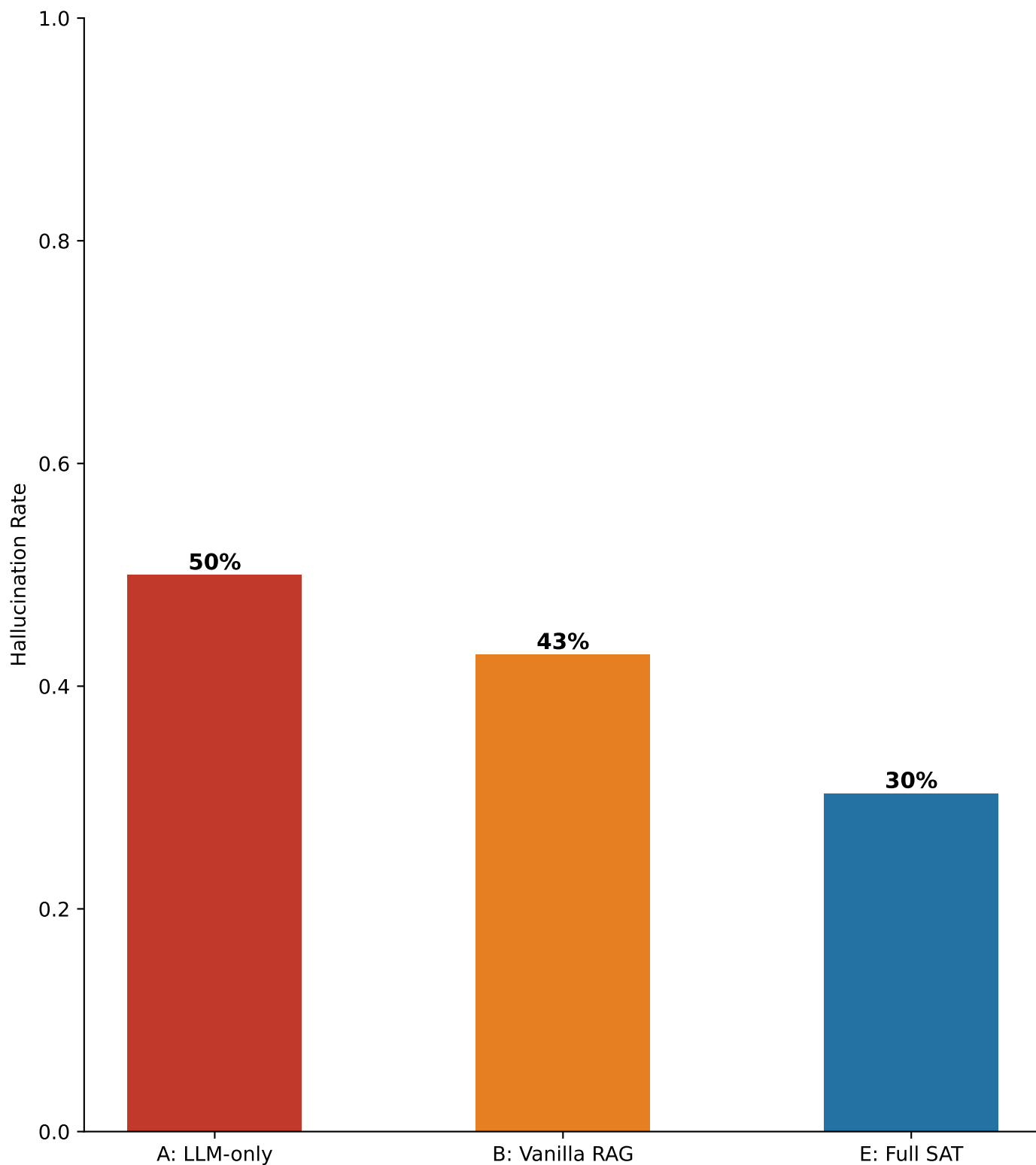
2c. Context Size / Pruning

How much raw retrieved context is discarded before generation

System	Avg. candidate sentences	Avg. kept	Reduction
B: Vanilla RAG	415	8	98.1%
E: Full SAT	377	8	97.9%

3. Hallucination Rate on Unsupported Questions

Objective measurement: corpus verified to contain ZERO relevant evidence -- any confident answer is fabricated (lower is better)

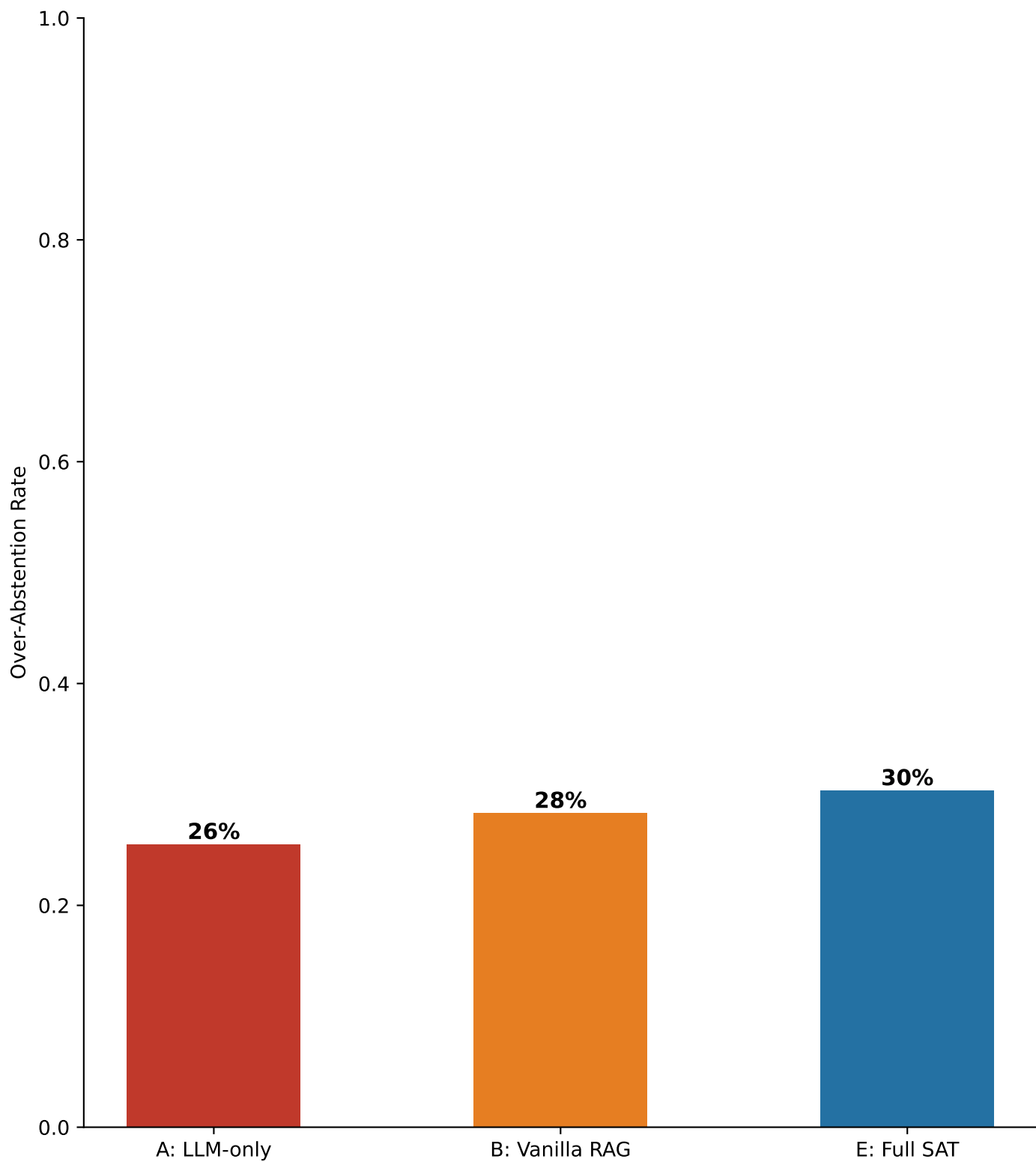


3b. Abstention Detail

System	N questions	Correctly abstained	Hallucinated
A: LLM-only	84	50.0%	50.0%
B: Vanilla RAG	84	57.1%	42.9%
E: Full SAT	84	69.6%	30.4%

3c. Over-Abstention on Answerable Questions

Refusing to answer when real evidence WAS available -- a different failure mode (lower is better)



3d. Fabricated Review Citations (System A only)

System A is given ZERO review text. Any response that says 'reviewers mention...' / 'customers say...' is inventing review content that was never shown to it.

System	N questions	Rate
A: LLM-only	476	98.3%

3e. Wrong-Brand Hallucination Rate

A brand mention that matches neither the product nor anything in the evidence shown to the model (a review citing a competitor phone for comparison doesn't count -- only a mention appearing from nowhere does). Rates are low across all systems; every one of Full SAT's 4 cases traces to the same Doorman CLARIFY bug in Section 5c, not to the answer-generation path.

System	N checked	Rate (count)
A: LLM-only	476	0.2% (1)
B: Vanilla RAG	476	0.0% (0)
E: Full SAT	476	0.8% (4)

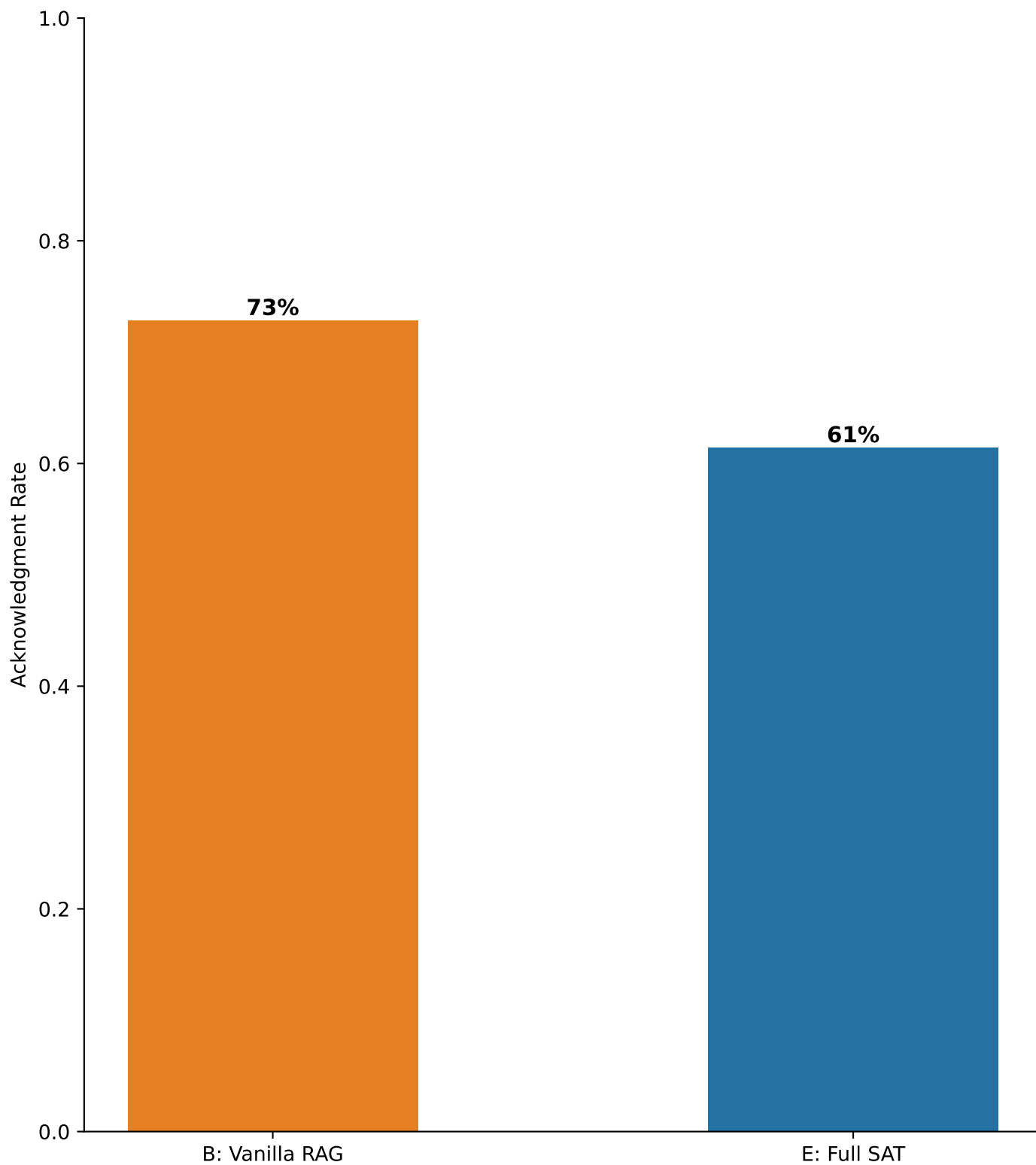
4. LLM-Judge Grounding Check -- Judge Was Unreliable

This did NOT come back clean -- the judge failed to produce valid structured output 20-36% of the time, and among responses it DID score, it almost never flagged anything as hallucinated. That pattern (high failure rate + near-zero discrimination) reads as the judge failing to do useful fact-checking, not as evidence of near-perfect grounding. Treat Section 3's objective measurements as authoritative; do not read this section as '0% hallucination'. See Limitations.

System	N attempted	Parse failures	Fail rate	N scored	Flagged hallucinated
B: Vanilla RAG	100	30	30%	70	0.0%
E: Full SAT	100	36	36%	64	1.6%

5. Contradiction Acknowledgment Rate

On queries with verified conflicting evidence (both positive AND negative reviews on the same aspect) -- does the response say so? (higher is better)



5b. Clarification Behavior (System E only)

Baselines A/B have no clarification mechanism -- they always attempt an answer

Metric	N	Value
Clarify recall (on genuinely ambiguous queries)	28	89.3%
False-clarify rate (on answerable queries)	476	1.7%

5c. Bug: Doorman Hallucinates Brand Names

Rare (under 2% of queries) but real: on a handful of unsupported-feature questions, Doorman's CLARIFY branch wrote a clarifying question that named a specific real phone (e.g. 'Samsung Galaxy S21 Ultra') unrelated to the actual product being discussed. Same root cause as the earlier interactive-CLI fix (echoing/rephrasing instead of asking); that fix reduced but did not eliminate this failure mode.

Metric	N	Count
False CLARIFY on non-ambiguous queries	476	8 (1.7%)
...of those, hallucinated a brand name in the clarifying question	8	4

6. Efficiency

Latency measured under batched generation (batch=8) -- approximate per-item cost, not single-request latency

System	p50 latency (s)	p95 latency (s)	Mean input tok.	Mean output tok.
A: LLM-only	1.48	1.77	97	115
B: Vanilla RAG	1.40	1.66	225	109
E: Full SAT	1.90	2.29	396	99

7. Findings & What Should Be Fixed Next

Hallucination on genuinely unsupported questions dropped from 50% (LLM-only) to 30% (Full SAT) -- a 20%-point reduction. This is the report's central claim (structured decomposition + an evidence-sufficiency gate reduces hallucination) and it holds on an objective, ground-truth-backed measurement, not a judge's opinion.

Against vanilla RAG specifically (same retrieval, no aspect-pruning/abstention gate), Full SAT meaningfully outperforms the baseline on unsupported-question hallucination (43% vs 30%). This isolates the value of the sufficiency gate + structured evidence, not just having retrieval at all.

System A fabricated review content it was never shown in 98% of responses -- confirming that without retrieval, the model defaults to writing AS IF it had read reviews, a distinct and important hallucination mode from simply guessing at specs.

Full SAT over-abstains (refuses to answer despite real evidence being available) on 30% of answerable questions -- worth investigating: is the evidence-sufficiency threshold too conservative? This is the retrieval-gate calibration issue flagged during development (PROJECT_NOTES.md) and should be tuned against the DEV set, not this frozen test set.

On queries with verified conflicting evidence, Full SAT acknowledged the disagreement 61% of the time vs 73% for vanilla RAG -- counter-intuitively, vanilla RAG acknowledged it MORE often -- the Analyst's structuring or the sufficiency gate may be suppressing contradiction signal here rather than helping; worth investigating rather than assumed as a win.

Doorman correctly triggered clarification on 89% of genuinely ambiguous queries, with a 2% false-clarify rate on answerable queries -- a capability the baselines don't have at all (they always attempt an answer, even to "is it good?").

7. Findings & What Should Be Fixed Next (cont.)

The secondary LLM-judge check did NOT produce a usable result: 30%/36% (B/E) of judgments failed to parse as valid structured output, and among the ones that did parse, essentially none were flagged as hallucinated (0%/1.6%). That combination reads as the judge failing to discriminate at all -- not as confirmation of near-perfect grounding. Using the same model family to judge its own outputs was not reliable here; a genuinely independent judge (a different model, or human review) is needed before this check is trustworthy. The objective measurements in Section 3 remain the credible hallucination numbers from this evaluation.

Full SAT's extra Doorman routing step costs latency: p50 1.9s vs 1.4s for vanilla RAG. Whether that's worth the hallucination reduction depends on the deployment's latency budget.

8. Limitations (1/2)

This is not human evaluation

No human annotators reviewed the gold evidence labels or the generated responses. The core hallucination metric (Section 3) is objective and doesn't need human judgment -- it relies on verified absence of evidence, not opinion -- but persona-alignment quality, answer completeness, and factuality beyond 'is it in the retrieved evidence' were not human-checked.

The LLM-judge is not independent

The grounding judge (Section 4) is the SAME base model (adapter disabled) that generated the responses being judged, not a different/stronger model or a human. A model judging outputs from its own weight family is a real source of bias -- treat Section 4 as a secondary, exploratory signal, and weight the objective measurements in Section 3 more heavily.

Retrieval-gate threshold not yet calibrated

The evidence-sufficiency gate (MIN_SCORE=0.55, MIN_EVIDENCE_SENTENCES=3) was carried over from training-data generation, not tuned against this dev/test split. The dev set (eval_dev.json) now exists specifically to do that calibration properly -- this report's numbers are the BEFORE, not the final, tuned result.

8. Limitations (2/2)

No full 5-tier ablation

The report's evaluation plan calls for retrieval / +pruning / +Analyst / +full-pipeline as separate tiers. This evaluation compares 3 systems (LLM-only, vanilla RAG, full pipeline), not all 5 -- the individual contribution of pruning vs Analyst vs abstention-gating separately is not isolated here.

Contradiction/clarify checks are keyword-heuristic

Contradiction acknowledgment and abstention detection use regex pattern matching over the response text, not semantic understanding -- a response that acknowledges disagreement in an unusual phrasing could be undercounted.

Single hardware run, no repeated trials

Latency figures are from one run, batched, on a single A30 GPU -- not averaged over multiple runs, and not representative of single-request (unbatched) production latency.